

Abhishek Sharma

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RESEARCH INTERESTS

Systems for large-scale AI/ML workloads — distributed and replicated systems (consensus, fault tolerance, linearizability), infrastructure for LLM serving and orchestration, and performance/reliability engineering for data-intensive and high-performance computing pipelines.

EDUCATION

- **IOE Pulchowk Campus, Lalitpur, Nepal** Nov. 2019 – Apr. 2024
Bachelor in Computer Engineering 76% average
- **SOSHGS, Gandaki, Pokhara, Nepal** 2017 – 2019
+2 Mathematics 3.85 GPA (CGPA 3.72)

PUBLICATIONS

Topic Modeling for Nepali Political News — *IEEE ICICT 2025* | [Paper](#) | [Code](#)
Compared probabilistic (LDA), algebraic (NMF, LSA), and transformer-based (BERTopic) methods on a 2018–2023 Nepali political news corpus; LDA produced more coherent topics than pretrained transformers in this low-resource language.

PROJECTS

- Quasar: Linearizable Replicated Key-Value Store** | [Link](#) | *Raft, SMR, WAL, FastAPI, Python*
- Implemented a Raft replicated state machine with term-based leader election, log-matching replication, majority-quorum commit, plus crash-durable WAL, snapshot compaction, and InstallSnapshot catch-up under crash and partition.
 - Served linearizable reads via a majority quorum check so an isolated leader cannot return stale state; built an interactive lab to step through election, partition, crash, and stale-leader recovery on a live three-node cluster.
- Videolize: Text Script to Video Generator** | [Link](#) | *Stable Diffusion, Gemini API*
- Script-to-video pipeline using Stable Diffusion, spherical latent interpolation, and Gemini API; 1st prize, GenAI category, Locus Hack-A-Week 2023.
- Information Extraction from Invoices using LLM** | *GPT-4 Vision, YOLO, Table-Transformers*
- Table and key-value extraction with GPT-4 Vision and fine-tuned Table-Transformers/YOLO; 1st prize, LLM category, Kathmandu University AI Competition 2023.
- Checkers** | [Link](#) | *C++, Minimax, Alpha-beta pruning*
- C++ board-game engine with Minimax and alpha-beta pruning as an AI opponent.
- Procedural Terrain Generator** | [Link](#) | *C++, Perlin noise*
- Hill terrain from Perlin noise, with controls for randomness, clipping, and wireframe rendering.
- University Network Infrastructure Design** | [Link](#) | *Cisco Packet Tracer*
- Campus topology with subnetting, VLANs, DNS/web servers, and OSPF dynamic routing in Cisco Packet Tracer.

PROFESSIONAL EXPERIENCE

- Founding Software Engineer** April 2025 – Present
ZeroTB San Francisco, California
- Developed AI-powered workflows using RAG and LLM orchestration frameworks (LangChain, LlamaIndex) for automated compliance document processing.
 - Built core backend infrastructure and APIs from the ground up, owning system architecture and scalability decisions in an early-stage startup environment.
 - Engineered the database schema, optimized database performance, and executed large-scale data migrations and third-party system integrations for production workloads.

- Collaborated across product and engineering to rapidly prototype and ship backend/AI-infrastructure features in a fast-paced startup setting.

Associate Software Engineer (ML)

April 2024 – April 2025

Docsumo

Chabahil, Kathmandu

- Performed critical performance analysis on the key-value extraction pipeline and reduced extraction latency by ~27% (7–8 seconds), significantly improving system throughput.
- Worked closely with the SRE team to improve reliability and scalability of distributed systems using GCP, Kubernetes, Docker, and Redis.
- Contributed to the core document-extraction pipeline for key-value and OMR extraction systems, including research, implementation, optimization, and refactoring of production APIs.
- Built and maintained scalable backend services and APIs using Flask and MongoDB, supporting high-volume document-processing workflows.

TEACHING EXPERIENCE

Instructor

2023

LOCUS Fellowship, Pulchowk Campus

- Co-designed curriculum and taught the mathematical foundations of machine learning (linear algebra, optimization, supervised algorithms) to 100+ students, including implementations from first principles.

Programming Instructor

2021 – 2022

IT Club and PDSC, Pulchowk Campus

- Taught Django web development to 250+ students (PDSC) and Git/GitHub workflows to 40+ students in an in-person IT Club workshop.

LEADERSHIP & MEMBERSHIP

President

Apr 2023 – Feb 2024

Society of Kaski Engineering Students (SOKES), Pulchowk Campus

- Led a student society of 200+ engineering students from Kaski District at Pulchowk Campus.

Secretary

Jan 2022 – May 2023

IT Club, Pulchowk Campus

- Department representative for Electronics and Computer Engineering; organized Git and design workshops and expanded outreach to students from other campuses.

AWARDS & ACHIEVEMENTS

Winner — LLM category in AI Competition organized by Kathmandu University [Dec. 2023] | [Link](#)

Winner — Generative AI category in LOCUS 2024 HackAWeek (Hackathon) [Jan. 2024]

Full Scholarship UnderGrad — **Rank 34** in IOE Examination (over 14000+ students) | [Link](#)

Rank 1 — Entrance Exam of Kathmandu University [Electronics Department rank] (over 10000+ students) | [Link](#)

Golden Jubilee Scholarship — Awarded By Embassy of India in Kathmandu, Nepal

Position Holder — National Level Math Competition (Budanilkantha Math League) | [Link](#)

Entrance Topper — High School examination out of over 3000+ students

College Topper — High School (Grade 12) Board Examination [**3.85 GPA**]

School Topper — SEE Level Examination [**3.8 GPA**]

CERTIFICATIONS

Machine Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Deep Learning Specialization — Coursera & DeepLearning.AI | [Link](#)

Annual Nepal AI School — NAAMII,Nepal | [Link](#)

AWS Academy Graduate - AWS Academy Cloud Architecting — AWS | [Link](#)

AWS Academy Graduate - AWS Academy Data Engineering — AWS | [Link](#)

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, Dart, HTML/CSS

Frameworks: Flask, FastAPI, Django, PyTorch, Node.js, Express.js, Flutter

Knowledge: Linux/Unix, Distributed Systems (Raft, SMR, linearizability), Computer Networks, Web Dev, Computer Vision (YOLO, CNN, ResNets, U-Net, etc.), Mathematical Domain (Linear Algebra, Probability and Statistics, etc.), NLP (RNNs, LSTMs), Transformer, DSA

Developer Tools: Git, Docker, Selenium, Postman, VS Code, Visual Studio, Cisco Packet Tracer

Libraries: gensim, spaCy, NLTK, scikit-learn, NumPy, Matplotlib, seaborn, Pandas

Databases: SQL (PostgreSQL, MySQL, SQLite), NoSQL (MongoDB)

Cloud Services: GCP, AWS (S3, EC2, Lambda, etc.)